

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Cent AI

Apr 2026 – Jun 2026

Software Engineering Intern: Python, Medical AI, ONNX, TinyGrad, Kafka, Redis, AWS, RDP

Bengaluru

- Built and maintained an **ETL pipeline** connecting to **Neorad** to fetch, transport, and prepare **MRI scans** for downstream segmentation and reporting workflows.
- Evaluated and integrated **multiple medical-image segmentation models** to automate **organ segmentation**, reducing examination turnaround time by **25%**.
- Integrated **medical LLMs** to verify **doctors' reports** and identify inconsistencies requiring clinical review.
- Built a dedicated **AI output evaluation system** for segmentation and report-verification results, making **model quality measurable, reproducible, and reviewable**.

Bentham AI

Dec 2025 – Feb 2026

Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built a resilient **Puppeteer and Node.js automation engine** for automating **legal and compliance workflows** around **MCA company-registration filings**, with API lookups, retries, session recovery, validation checks, and operator checkpoints; reduced manual filing time by **85%**.
- Implemented **structured draft generation** for company-name suggestions, descriptions, and filing fields, with **inspectable outputs and deterministic validation** before execution.
- Containerized services with **Docker**, deployed them on **Google Cloud Run**, and automated releases with **GitHub Actions**, improving deployment velocity by **70%**.

Google Summer of Code

May 2025 – Sept 2025

Mentee at Invesalio: Python, VTK, NumPy, PyTorch, ONNX, TinyGrad

Remote

- Built **medical-imaging segmentation workflows** for **95 anatomical brain regions**, integrating **MRI preprocessing, PyTorch and TorchScript, ONNX and TinyGrad** inference, generated masks, and 3D inspection paths.
- Worked with **NIFTI and MGZ volumes**, voxel conformation, **orientation-sensitive preprocessing**, label remapping, LUT mapping, binary masks, and **QC checks** for model output correctness.
- Reduced MRI-processing runtime by **87%** through **asynchronous and parallel execution** while validating orientation, labels, **GPU and CPU** paths, and reproducible outputs.

Open Source

[Contributions](#)

- Contributed maintainer-reviewed patches across **Invesalio, Kuadrant MCP Gateway, Tiled, CRIU, and VideoLAN**, spanning medical-data correctness, backend failover, parser robustness, security validation, and compatibility fixes.

Projects

Hyoka Agent Reliability Layer | Python, FastAPI, SQLAlchemy, PostgreSQL, OpenTelemetry

[GitHub](#)

- Solved unreliable AI-agent behavior without visibility, evaluation, or rollback by building Hyoka, a **reliability control plane** for traces, workflow replay, failure mining, evaluations, and gated promotion.
- Enabled a **self-improvement loop** that converts failures into candidate changes, validates them against prior behavior, and promotes only gated improvements; stored artifacts, results, audit logs, and release metadata in **SQLite and PostgreSQL**.

Postificus Content Distribution Engine | Go, Echo, React, PostgreSQL, Redis, RabbitMQ, Go-Rod

[GitHub](#)

- Solved fragmented publishing across Medium, Dev.to, and LinkedIn with a **centralized Go engine** built around **Go-Rod browser automation** and AI-driven content transformation.
- Engineered distributed execution for long-running browser jobs with **RabbitMQ retries and dead-letter queues, Redis autosave, PostgreSQL persistence**, circuit breakers, health checks, and Prometheus metrics to isolate third-party failures.

Technical Skills

Languages: Python, Go, TypeScript, JavaScript

Frameworks: FastAPI, Next.js, Node.js, SQLAlchemy, PyTorch, ONNX

Tools: PostgreSQL, Redis, RabbitMQ, Docker, GCP, OpenTelemetry, Prometheus, GitHub Actions, REST APIs